

IA Mathematics for Natural Sciences A:

Examples Sheet 4 - Complex loci, complex exponentials and logarithms, and hyperbolic functions

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Sketching complex loci

1. By setting $z = x + iy$, identify and sketch the loci: (a) $\operatorname{Re}(z^2) = \operatorname{Im}(z^2)$; (b) $\operatorname{Im}(z^2) = -iz^2$.
2. (Circles, lines, and half-lines) Without setting $z = x + iy$, sketch the loci:
(a) $|z - 2 - i| = 6$; (b) $|z^* - i| = 1$; (c) $|z - 2| = |z + i|$; (d) $\arg(z) = \pi/2$; (e) $\arg(z^*) = \pi/4$.
If z lies on the locus in part (a), what are the loci of the complex variables $u = z + 5 - 8i$, $v = iz + 2$, and $w = \frac{3}{2}z + \frac{1}{2}z^*$?
3. (a) Show that $|z + k|^2 = zz^* + kz^* + k^*z + kk^*$.
Now consider the locus $azz^* + bz^* + b^*z + c = 0$, where $a, c \in \mathbb{R}$ and $b \in \mathbb{C}$ are constants.
(b) Without setting $z = x + iy$, use Question 1 to show that the locus is a line when $a = 0$.
(c) Without setting $z = x + iy$, use (a) to show that the locus is a circle, point, or empty, when $a \neq 0$.
(d) Hence, without setting $z = x + iy$, sketch the loci: (i) $z + z^* - 3 = 0$; (ii) $|z - 2| = 3|z - 1|$.
(e) (*) Generalising part (d)(ii), prove that the locus $|z - \alpha| = \lambda|z - \beta|$ is always a circle when $\lambda \neq 1$, and find its centre and radius. [*This form of a circle is called the Apollonian form of a circle.*]
4. (Two conic sections) Sketch the loci: (a) $|z - 4| + |z^* - i| = 3$; (b) $|z - 4| - |z^* - i| = 3$.

Exponentials, logarithms, and powers

5. State Euler's formula for the complex exponential $e^{i\theta}$. Hence, express each of the following complex exponentials in Cartesian form: (a) $e^{-i\pi/2}$; (b) $e^{-\pi}$; (c) $e^{i\pi/4}$; (d) e^{1+i} ; (e) $e^{2e^{i\pi/4}}$.
6. Find the real and imaginary parts, and the modulus and argument, of the expression $e^{i\omega t}/(R + iX)$, where ω, t, R and X are real. If you're taking Part IA Physics, what might these constants represent?
7. The complex logarithm, $\log(z)$, is defined to satisfy $e^{\log(z)} = z$ for all z .
(a) By setting $\log(z) = u + iv$, where u, v are real, find u and v in terms of $|z|$ and $\arg(z)$. Hence show that there are infinitely many possible values of the complex logarithm.
(b) Using the complex logarithm, find all complex numbers z satisfying: (i) $e^{2z} = -1$; (ii) $e^{z^*} = i + 1$.
8. The complex power z^w is defined by $z^w := e^{w\log(z)}$.
(a) By considering the cases when w is an integer, rational, irrational, or complex number separately, explain when complex powers are multi-valued.
(b) Find all possible values of: (i) $i^{1/3}$; (ii) i^i ; (iii) $i^{(i^i)}$; (iv) $(i^i)^i$.

Trigonometry with complex numbers

9. State and prove De Moivre's theorem. Use it to solve the equation $16 \sin^5(\theta) = \sin(5\theta)$.
10. Using Euler's formula, show that $\sin(\theta) = \frac{1}{2i}(e^{i\theta} - e^{-i\theta})$, and $\cos(\theta) = \frac{1}{2}(e^{i\theta} + e^{-i\theta})$. Hence express $\sin^5(\theta)$ in terms of $\sin(\theta), \sin(3\theta), \sin(5\theta)$.
11. Find all solutions of the equation $\cos(z) = 2$ in terms of logarithms.
12. Let $\theta \neq 2p\pi$ for $p \in \mathbb{Z}$. Show that $\sum_{n=0}^{N-1} \cos(n\theta) = \frac{\cos((N-1)\theta/2) \sin(N\theta/2)}{\sin(\theta/2)}$. What if $\theta = 2p\pi$?

Hyperbolic functions

13. State the definitions of $\sinh(x), \cosh(x)$, and hence show that $\sinh(ix) = i \sin(x), \cosh(ix) = \cos(x)$. Using this property, explain how a trigonometric identity may be transformed into a hyperbolic identity. Thus, give the hyperbolic analogues of the formulae:
$$\cos^2(x) + \sin^2(x) = 1, \quad \cos(x+y) = \cos(x)\cos(y) - \sin(x)\sin(y), \quad \tan(x+y) = \frac{\tan(x) + \tan(y)}{1 - \tan(x)\tan(y)}.$$
14. Express $\sinh^{-1}(x), \cosh^{-1}(x)$ and $\tanh^{-1}(x)$ as logarithms, justifying any sign choices you make.
15. Find all solutions to the equation $\cosh(z) = i$.